WASTEWISE-ML · MARKET AND VIABILITY

WasteWise-ml market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First market and viability note (kit 1.8 upgrade)

WasteWise-ml market and viability

WasteWise-ml is for waste picker cooperatives and the people in them, who sort by hand and earn by grade, with nothing between their own judgment and a plastics identifier that costs about EUR 4,930. It competes on being free software on a phone users already own, at about USD 183 for the rest of a station, and it is open so the cooperatives that co-own its data can keep and change it. Prices and benchmarks are cited with the date they were checked. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Waste picker cooperatives and depots	Member-owned cooperatives that sort dry waste for scrap dealers; first candidate SWaCH, Pune, India	Handheld NIR plastic identifier: Matoha PlasTell Handheld, EUR 4,930 (filament2print, checked 2026-10-25); identifies resin, not buyer grade or non-plastics	313.39 a station with a USD 130 reference phone; about 183 with members' own phones (BOM lines 2 to 7, 11, 12)	Class, local grade and hazard flag on a phone, offline; "Unsure" rather than a guess; grade names set by the cooperative
Individual waste pickers	Pickers in the street or at depots who already carry a low-cost Android phone	Low-cost phone of the class: Redmi A5, 3 GB and 64 GB, INR 6,499 at launch (The Statesman, 2025-04-17; checked 2026-10-25)	0 for the app on their own phone (handheld use)	Learn new packaging and grades; fewer disputes with buyers
Municipal drop-off points and material recovery facilities	Hand-sorting lines and drop-off staff in Indian cities	Industrial NIR sorters (price not published; hundreds of thousands of dollars per WasteWise Scan WSC-PRB-001, an estimate)	313.39 a station	A quality check on hand-sorted lines and a battery flag before baling
Schools, campuses and events	Teaching which bin an item belongs in	No paid benchmark checked	0 to 183	Teaching tool with the "Unsure" route shown openly
Researchers and NGOs	Universities and programs extending the model to new regions	Public datasets, free (TrashNet MIT; TACO CC BY 4.0 annotations)	Labeling effort: USD 690 for the dataset and evaluation set at the planning rate	Open model, taxonomy, evaluation method and consent protocol to reuse

Option B, the tray detector, would cost about USD 419 a station without a phone, from a Raspberry Pi 5 4 GB at USD 100 (after the February 2026 rise, OMG! Ubuntu, 2026-02-02), a Camera Module 3 Wide at USD 35.95 and a 5 inch Touch Display 2 at USD 54.95 (PiShop.us, checked 2026-10-25), plus estimates (WML-CAL-001 section 15). The reference phone of option A, priced at USD 130 in the BOM as an indicative figure, sits above the Redmi A5's launch price, so a cheaper qualifying phone may exist; the price is confirmed when the phone is bought (WML-DEC-001 confirm item 1).

The benefit to a picker is the price difference between a sorted grade and mixed material, times the mass moved up a grade. No local price list has been collected yet, so no figure is claimed; the first co-design sessions should record the buyer's price list and the mass a picker sorts a day.

Open route

- **Who builds it.** The cooperative's own members or a local workshop build the stand from the build plan (saw, drill, files); every other station item is bought locally. The app installs on members' phones.
- **How it spreads.** Through waste picker networks (WIEGO and the International Alliance of Waste Pickers), the Helpful Engineering volunteer network, universities and city programs; a new city needs a new grade file, not a new model, unless it pays for a grade the model cannot see.
- **What keeps it open.** Code and weights are MIT; every dataset license is recorded; field data is co-owned by the pickers' organization (D6); the evaluation method and model card are published with every release. No paid cloud service is needed at the point of use.
- **Non-commercial by intent.** The project sells nothing; any income from sorted material stays with the pickers.

Sources

- filament2print. "Identificador de Plásticos Matoha PlasTell" (PlasTell Handheld EUR 4,930.00). https://filament2print.com/es/fabricacion-filamentos/4664-identificador-plasticos-matoha-plastell.html. Checked 2026-10-25.
- OMG! Ubuntu. "Raspberry Pi Announce More Price Rises." 2026-02-02. https://www.omgubuntu.co.uk/2026/02/raspberry-pi-annouce-more-price-rises. Checked 2026-10-25.
- PiShop.us. Raspberry Pi category, page 2. https://www.pishop.us/raspberry-pi-foundation/?page=2. Checked 2026-10-25.
- The Statesman. "Xiaomi Launches Redmi A5 in India, Price Starts at Rs 6,499." 2025-04-17. https://www.thestatesman.com/technology/xiaomi-launches-redmi-a5-in-india-price-starts-at-rs-6499-1503421119.html. Checked 2026-10-25.
- WasteWise Scan, WSC-PRB-001 (industrial NIR sorter cost, an estimate there).
- BOM `bom/bom.csv` and WML-CAL-001 sections 10 and 15 for every cost of this design.